

Shadman Salif Swanan

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EDUCATION

BRAC University

Bachelor of Science in Computer Science and Engineering (CSE)

Jan. 2022 – Present
Dhaka, Bangladesh

- **CGPA:** 3.82 out of 4.00
- **Programming Coursework:** Fundamentals of Programming, Object-Oriented Programming, Data Structures & Algorithms
- **Computational Coursework:** Natural Language Processing, Artificial Intelligence, Computer Network, Linear Algebra, Neural Networks, Digital Logic Design, VLSI Design, Discrete Mathematics
- **Programming Languages:** Python, Java, C/C++, JavaScript, TypeScript
- **Frameworks & Tools:** PyTorch, OpenCV, NumPy, Matplotlib, Scikit-learn, Git, OpenGL, MySQL, React, Next.js, Figma, Cisco Packet Tracer

Notre Dame College

Higher Secondary Certificate (HSC), Science

2018 – 2021
Dhaka, Bangladesh

- **GPA:** 5.00 out of 5.00

Ideal School and College

Secondary School Certificate (SSC), Science

2008 – 2018
Dhaka, Bangladesh

- **GPA:** 5.00 out of 5.00

PROJECTS

[Token-Level POS Tagging using RNN Models](#) Implemented POS tagging using RNN, LSTM, GRU, and BiLSTM architectures

[Disease Prediction using Machine Learning](#) Developed classification models for diabetes prediction using KNN, Decision Tree, and Naive Bayes

[Shikkha.com](#) Developed an educational platform using JavaScript for digital learning support

[Federal Flight Agency Mission-Critical Aerospace Network Design](#) Designed a secure and scalable aerospace network architecture for mission-critical communication systems

[3D Car Racing Game](#) Built a 3D racing game using OpenGL with enemy AI, shooting mechanics, weather effects, and camera modes

[Automated Sanitizer and Mask Dispensing Robot](#) Embedded systems project integrating automated sanitizer dispensing and mask detection

RESEARCH

Undergraduate Thesis Research

BRAC University

Sept. 2025 – Present

- Conducting research on **semantic drift in the Bengali language** using diachronic word embeddings and transformer-based representations.
- Developing a column detection model for scanned Bengali newspapers to improve document layout understanding.

- Building an OCR pipeline to construct a clean and high-quality diachronic text corpus for downstream NLP tasks.
- Performed **Exploratory Data Analysis (EDA)** to quantify OCR error distributions and validated post-processing techniques that reduced linguistic noise across a 120-year diachronic corpus
- Fine-tuned a **Historical BanglaBERT** model on a curated diachronic corpus and analyzed semantic drift using contextual embeddings.

Leaf Disease Research (TinyResNet)

April 2026

Independent Research Project

- Developed a lightweight deep learning model (TinyResNet) for tomato leaf disease classification under real-world conditions
- Proposed **distortion-aware augmentation** (blur, noise, low-light) to improve model robustness
- Conducted comparative analysis with ResNet18 baseline, achieving **96.59% accuracy** and superior generalization
- Evaluated model performance using confusion matrix, precision, recall, and F1-score across multiple disease classes

Real-Time Fraud Detection System

May 2026

Independent Research Project

- Engineered an end-to-end ML pipeline using **LightGBM** to identify fraudulent credit card transactions within a highly imbalanced dataset (0.17% fraud rate).
- Implemented **cost-sensitive learning** via scale-pos-weight and threshold tuning to optimize the precision-recall trade-off, ensuring detection of high-risk "micro-charges."
- Developed a live monitoring console in **Python** that simulates real-time transaction streams and triggers instant security alerts based on model probability scores.
- Performed extensive **Exploratory Data Analysis (EDA)** using Seaborn and Matplotlib to visualize feature correlations and isolate anomalies in PCA-transformed transaction data.

EXPERIENCE

University of Tokyo (Matsuo-Iwasawa Laboratory)

Apr. 2026 – Present

GCI World Program (Global Consumer Intelligence)

- Participating in an international AI education program focused on data science and machine learning
- Engaging with global participants to develop practical and theoretical understanding of AI applications

Grameenphone Academy

Mar. 2026

UX Design Decision-Making with AI & Design System Thinking with AI

- Completed certifications focusing on AI-assisted user-centered design and structured design system thinking
- Explored how AI enhances product design, decision-making, and scalable system development

Notre Dame College Model United Nations

Conference Management Officer

- Managed responsibilities in the Conference Management Department during a large-scale academic event
- Demonstrated strong organizational skills, adaptability, and ability to work efficiently under pressure

CO-CURRICULAR ACTIVITIES

Notre Dame Nature Study Club

Vice President, Department of Photography

- Led photography initiatives and coordinated visual documentation for club activities and events
- Managed team members and contributed to creative direction and event coverage

BRAC University Residential Semester

Stage Drama Course Participant

- Completed an intensive stage drama course emphasizing teamwork, communication, and performance skills

BRAC University Sports Activities

Football Achievements

- Champion & Top Scorer, Intra-Spring 2022 Football Tournament, BRAC University
- Runners-Up, Crusade 2026 (Semester Tournament)
- Runners-Up, Residential Semester (RS) Football Tournament
- Runners-Up, RS Cup